



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
19-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

Time : 3 Hours

PAPER-1

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

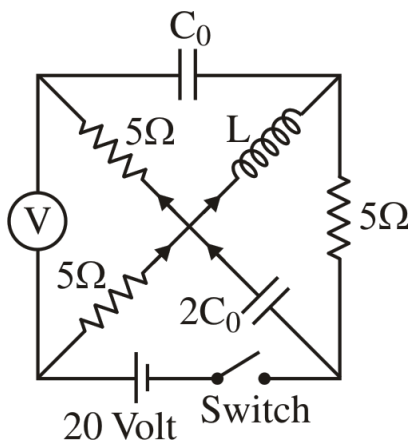
- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

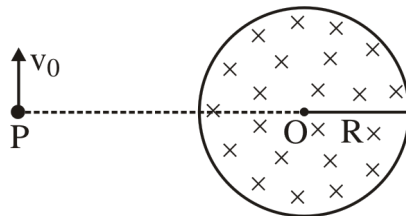
1. For the given LCR circuit, if voltmeter is ideal and switch was initially in closed state for very long interval of time, find reading of voltmeter just after opening the switch. (Given : $C_0 = 2\mu\text{F}$, $L = 1$ henry, $R = 5\Omega$, $V = 20$ volt)



- | | |
|-------------|-------------|
| (A) Zero | (B) 10 Volt |
| (C) 20 Volt | (D) 30 Volt |

Space for Rough Work

2. In the cylindrical region of radius $R = 2\text{m}$, there exist a time varying magnetic field B which increases at a rate of $\frac{dB}{dt} = 4 \text{ Tesla/sec}$. A charged particle having charge $q = 2\text{C}$ is placed at the point P at a distance d from its centre O . Now, the particle is moved in the direction perpendicular to the line OP by an external agent up to infinity. So that there is no gain in kinetic energy of charged particle distance between OP is 4m .

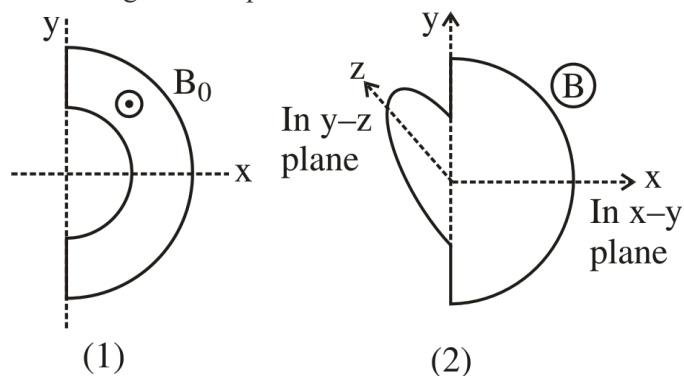


$R = 2\text{m}; OP = 4\text{m}$

- (A) $2\pi \text{ Joule}$ (B) $4\pi \text{ Joule}$
(C) $6\pi \text{ Joule}$ (D) $8\pi \text{ Joule}$

Space for Rough Work

3. The drawing shows a coil of copper wire that consists of two semicircles joined by straight section of wire. In part (a) the coil is lying flat on a horizontal surface. The dashed line also lies in the plane of the horizontal surface starting from the orientation in diagram (1). The smaller semicircle rotates at an angular frequency ' ω ' about the dashed line until, its plane becomes perpendicular to the horizontal surface as shown in diagram (2). A uniform magnetic field B constant in time and is directed upward perpendicular to the horizontal surface. The magnitude of magnetic field is $B = 2\text{ T}$. The resistance of the coil is $0.4\ \Omega$ and the smaller semicircle has a radius of 20 cm while the bigger one has a radius of 40 cm . The angular frequency at which the small semicircle rotates is $\omega = 15\text{ rad/sec}$. Determine average current I , if any induced in the coil as the coil changes its shapes



(A) 2 amp

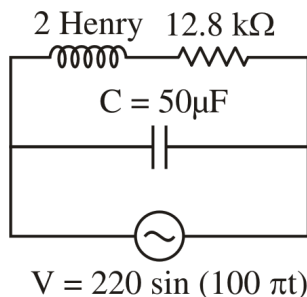
(B) 0 amp

(C) 1 amp

(D) 3 amp

Space for Rough Work

4. Choose the correct option regarding the following AC circuit.



- (A) Resonance occurs when 100 Hz.
 (B) Resonance occurs when power factor is minimum.
 (C) Resonance occurs when reactive power of circuit becomes equal to resistance of wire.
 (D) Resonance occurs at 60 rad/sec.
5. In a moving coil galvanometer the number of turn $N = 50$, area of the coil $A = 8 \times 10^{-3} \text{ m}^2$ and the magnetic field strength $B = 0.5 \text{ Tesla}$. Torsion constant of spring is $40 \times 10^{-3} \text{ SI unit}$ and resistance of a coil is 2.5Ω . Then the voltage sensitivity of moving coil galvanometer (in SI unit) is :-
 (A) 1 (B) 2 (C) 3 (D) 4
6. A particle is subjected to two simple harmonic motions along the x and y axes, described by $x(t) = a \sin(2\omega t + \pi)$ and $y(t) = 2a \sin(\omega t)$. The resultant motion is given by
 (A) $\frac{x^2}{a^2} + \frac{y^2}{4a^2} = 1$ (B) $x^2 + y^2 = 1$
 (C) $y^2 = x^2 \left(1 + \frac{x^2}{4a^2}\right)$ (D) $x^2 = y^2 \left(1 - \frac{y^2}{4a^2}\right)$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

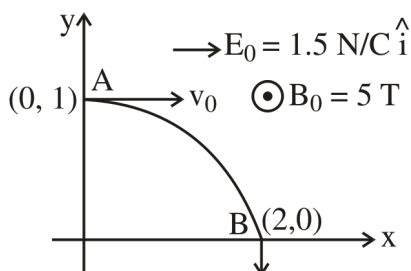
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. A mass of 10kg falls 50 cm on to the platform of a spring scale and sticks. The platform which eventually comes to rest "10 cm" below the initial position. The mass of the platform is "2kg". Then select the correct statement :

- (A) Spring constant of the spring is approximately equal to 980 N/m.
- (B) Spring constant of the spring is approximately equal to 490 N/m.
- (C) If we put a damping system so that the scale comes to rest in minimum time (i.e. critical damping) then damping constant is 18 sec^{-1} .
- (D) If we put a damping system so that the scale comes to rest in minimum time (i.e. critical damping) then damping constant is 9 sec^{-1} .

Space for Rough Work

8. A particle is projected with a speed of 2 m/s from (0, 1) in a region of magnetic field of 5 Tesla pointing in z direction. If particle crosses the x-axis at (2, 0) along the negative y direction. Electric field in the given space 1.5 N/C along positive x-direction. Mass of charged particle is 100 gm and the charged particle has charge of magnitude 0.2 C. Select the correct statement.



- (A) Velocity with which charged particle crosses x-axis is 4 m/s.
 (B) Time taken by particle from point A to B is $8/3$ sec.
 (C) Radius of curvature of particle at point B is 2 m.
 (D) Velocity of particle when it crosses point B is 3 m/s.
9. The self inductance of coil 'b' is twice that of coil 'a'. When they are connected as shown in figure-1, net induction is 120 mH. When they are connected as in figure-2 the net induction is 60 mH. Then select the correct statement(s) from the following :

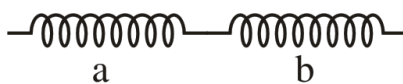


Figure-1

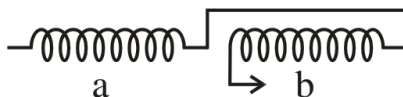
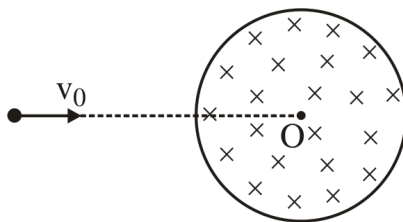


Figure-2

- (A) $M_{12} = 20$ mH (B) $M_{12} = 30$ mH (C) $L_A = 60$ mH (D) $L_A = 30$ mH

Space for Rough Work

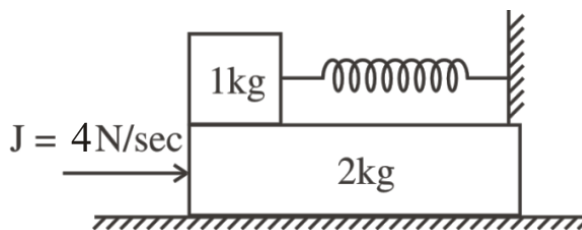
10. An electromagnetic radiation is emanating from a 100 watt lamp. Assuming that 20% of the power consumed is radiated as light and point 'P' is at a distance of 1m from the lamp then select the correct statement(s).
- (A) Electric field peak at point 'P' is 34.7 V/m.
- (B) Electric field peak at point 'P' is nearly $6\sqrt{2} \times 10^5$ V/m.
- (C) Magnetic field peak at point 'P' is nearly 1.2×10^{-7} tesla.
- (D) Magnetic field peak at point 'P' is nearly 2.88×10^{-7} tesla.
11. Uniform magnetic field is confined into a cylindrical region. If α -particles with different speed enter the region radially, then :-



- (A) Faster the particle, lesser is the time spent in the region.
- (B) Slower the particle, lesser is the time spent in the region.
- (C) Slower the particle, greater is the time spent in the region.
- (D) Time will be same for all particles.

Space for Rough Work

12. A block of mass $m = 1 \text{ kg}$ is placed on a plank of mass $= 2 \text{ kg}$, which is placed on a horizontal frictionless surface. There is no friction between the block and the plank. Block and plank are connected by a spring of spring constant 6 N/m as shown in the figure. An impulse $J = 4 \text{ Newton/sec}$ is applied on the plank. Possible magnitude of acceleration of the block are :



- (A) 1 m/s^2 (B) 3 m/s^2
(C) 5 m/s^2 (D) 7 m/s^2

Space for Rough Work

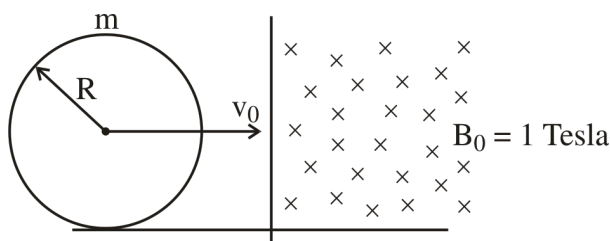
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

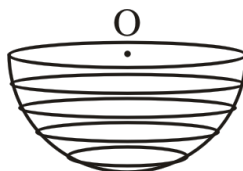
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A ring of mass 1 gm and radius 10 mm is made of an insulating material carries uniformly distributed charge. Initially it is moving on a frictionless horizontal table top with its plane vertical with velocity $\sqrt{2}$ m/s. The charge on the ring such that it starts rolling on entering completely into the region of the magnetic field is $\sqrt{N}$. Then find the value of N.

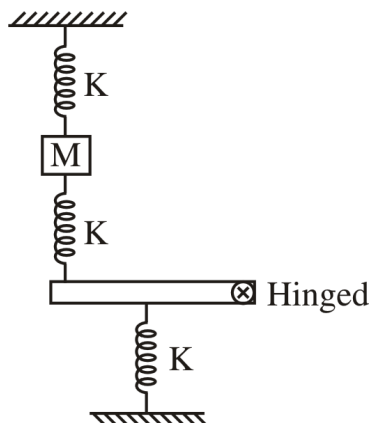


2. A wire is wrapped over a hemi-spherical shell of radius 'R'. If N = number of turns and I = current per turn and the induction at O is $\frac{\mu_0 NI}{nR}$, the value of n is :-



Space for Rough Work

3. What will be angular frequency (rad/sec) for small oscillation of block of mass m in the arrangement shown ? Neglect the mass of the rod, where K is 100 N/m and mass of block is 30 kg, rod is massless and hinged at 'O' as shown in diagram.



4. In an LCR series circuit, a non-inductive resistor of 150Ω , a coil of $\frac{35}{44}$ H inductance and negligible resistance, and a $\frac{700}{22}\mu\text{F}$ capacitor are connected across an ac power source of 220V, 50 Hz. The power loss across the resistor is _____ W (Round off a 2 decimal places).
5. In a region of space, a time dependent magnetic field $B(t) = 0.4t$ tesla points vertically upwards. Consider a horizontal, circular loop of radius 2 cm in this region. The magnitude of the electric field (in milli volt /meter) induced in the loop is.....
6. Two parallel rails of a railroad track are insulated from each other and from the ground. The distance between the rails is 1 meter. A voltmeter is electrically connected between the rails. Assume the vertical component of the earth's magnetic field to be 0.2 gauss. What is the voltage developed between the rails when a train travels at a speed of 180 km/h along the track? Give the answer in milli-volts. [Given 1 Gauss = 10^{-4} Tesla]

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
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- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

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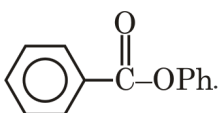
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Which of the following statement is **INCORRECT** ?

(A) Adenine which is part of DNA and RNA both, give positive carbyl amine test.

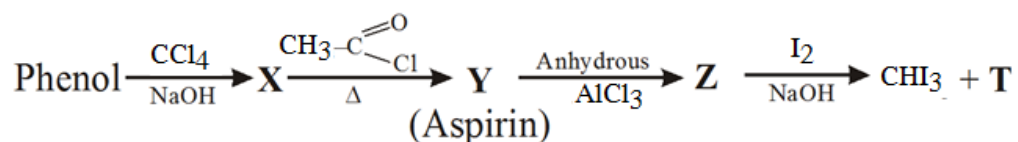
(B) Aspartame has 9 sp^2 hybridised carbon atom and 2 chiral centre

(C) Electrophilic aromatic substitution takes place in deactivated ring in 

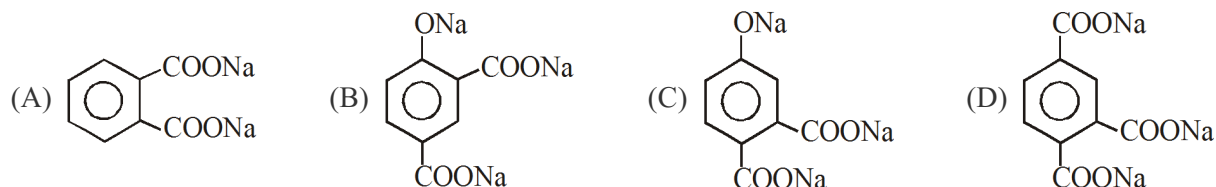
(D) Chlorobenzene has lower rate of electrophilic aromatic substitution than benzene

Space for Rough Work

2. Consider given reaction sequence-

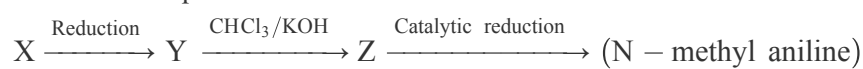


Product 'T' is



3. Which of the following is incorrect ?

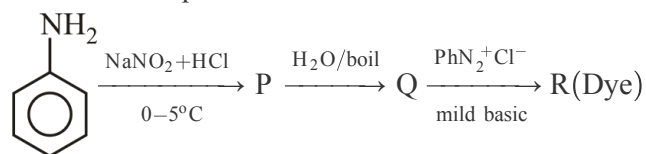
(A) In reaction sequence



compound X can be nitrobenzene

(B) Compound  can not undergo hell-volhard-zelinsky reaction.

(C) In reaction sequence

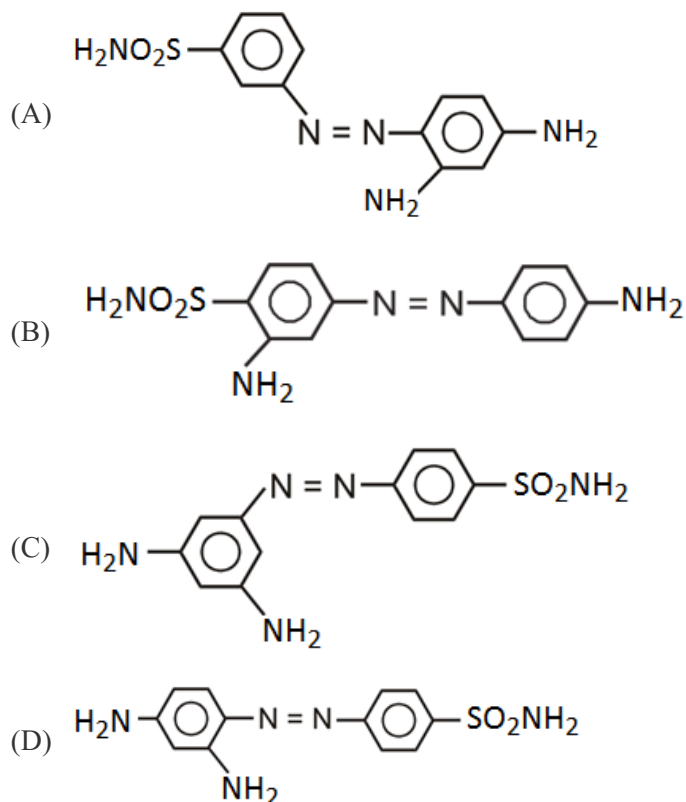


product R is having red/orange color.

(D) Glucose gives +ve 2,4-DNP test.

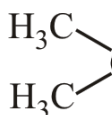
Space for Rough Work

4. An aqueous solution of starch having concentration 10.0 g/L has an osmotic pressure 5.0×10^{-3} atm at 25°C. Then **approximate number of glucose units** in this sample of starch will be....
- (A) 300 (B) 302 (C) 304 (D) 306
5. Coupling reactions can be used in synthesis of dyes and in synthesis of various other compounds. There is a drug sulphonamide also known as prontosil. It is prepared by diazotising sulphanilamide (p-Amino benzene sulphonamide) and then its coupling with benzene 1,3-diamine. The structure of prontosil is:

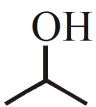
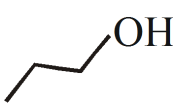


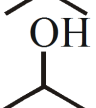
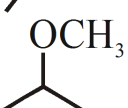
Space for Rough Work

6. Consider following statement(s)-

(i) $\text{CH}_3\text{CH}_2\text{NH}_2$ &  $\text{CH}-\text{NH}_2$ can be differentiated by Hinsberg test.

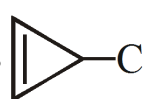
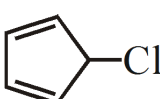
(ii) Both PhCH=O and PhCOCH_3 in presence of NaOH undergo Cannizzaro reaction.

(iii) Compounds  &  can be differentiated by $\text{NaOH} + \text{I}_2$.

(iv) Compounds  &  can be differentiated by using ceric ammonium nitrate test.

(v) 2-hexanone and 3-hexanone can be differentiated using NaHSO_3 .

(vi) Sucrose has both hemiacetal and acetal linkage.

(vii) Compounds  &  can be differentiated by aqueous AgNO_3 .

(viii) Acetaldehyde and Benzaldehyde can be differentiated by Tollen's test.

How many of these given statements are correct ?

(A) 6

(B) 7

(C) 4

(D) 8

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. An organic compound (X) having five carbon atoms on boiling with NaOH gives two compounds (Y) and (Z). Y on acidification gives (T) which turns blue litmus paper to red. (Z) gives +ve iodoform test. Ag salt of (T) contains 59.66% Ag. Which of the following option(s) is/are correct

(A) Structure of (X) can be $\text{CH}_3\text{--CH}_2\text{--COO--CH}_2\text{--CH}_3$

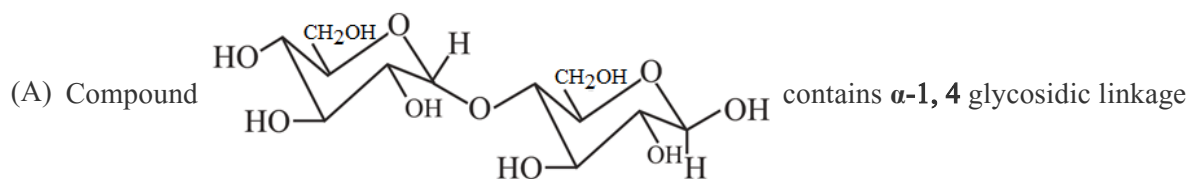
(B) Structure of (T) can be $\text{CH}_3\text{--CH}_2\text{--COOH}$

(C) Sodium salt of acid (T) can be $\text{CH}_3\text{--CH}_2\text{--COONa}$

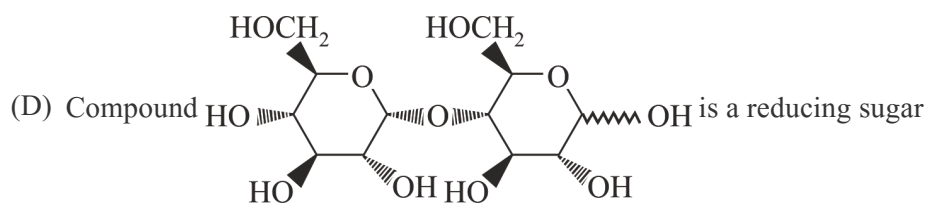
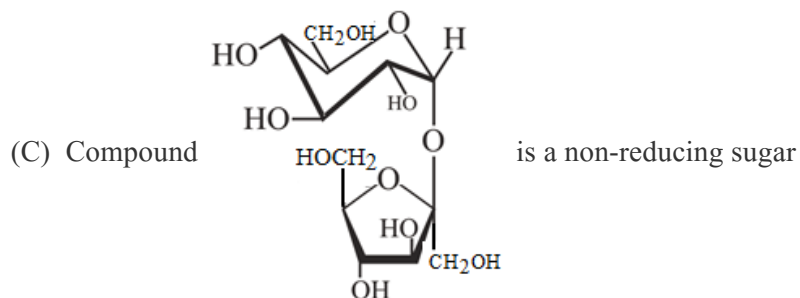
(D) T is valeric acid

Space for Rough Work

8. Which of the following statement(s) is/are correct ?

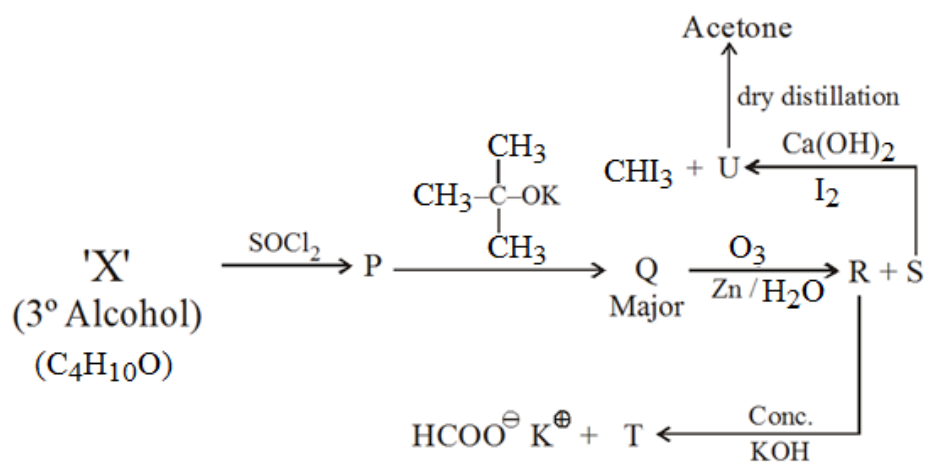


(B) Lactose is example of non-reducing disaccharide.



Space for Rough Work

9. Consider following set of reactions :



Which of the following statement(s) is/are correct ?

- (A) Unknown alcohol 'X' is 
- (B) R & S can be distinguished by Fehling solution
- (C) Heat of combustion of compound Q is greater than But-2-ene
- (D) Reaction of compound R with conc. KOH involves Cannizzaro reaction

Space for Rough Work

10. X, Y and Z are three compounds having same empirical formula CH_2O . Given that molecular weight of Y is 60 and molecular weight of Z is also 60. Molecular weight of X is 30.
X gives (+ve) Tollens and (+ve) Fehling test but gives (–ve) NaHCO_3 test.
Y gives (–ve) Tollens and (–ve) Fehling test but gives (+ve) NaHCO_3 test.
Z gives (–ve) NaHCO_3 test and having fruity smell.
Which of the following statement(s) is/are correct about X, Y and Z ?
- (A) Compound X can give aldol condensation but not Cannizzaro reaction.
(B) Compound Y can give methane gas upon treatment with sodalime/ Δ .
(C) Compound Z upon reduction with LiAlH_4 gives only one type of alcohol.
(D) Compound Z is ester
11. Which of the following compound(s) on reaction with red P + Br_2 followed by reaction with NH_3 gives naturally occurring α -amino acids ?
- (A) HCO_2H (B) $\text{H}_2\text{C}=\text{O}$ (C) $\text{CH}_3\text{CO}_2\text{H}$ (D) $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
12. Which of the following statement(s) is/are not INCORRECT about amino acid(s) ?
- (A) Tyrosine, Tryptophan, Histidine and Proline contain ring(s).
(B) Arginine is having highest pH at isoelectric point among natural α -amino acids.
(C) Amino acids exist in form of zwitter ions.
(D) Threonine has 2 chiral centre

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

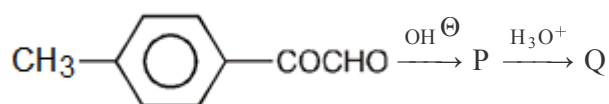
- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If x = DOU of **Aspirin**
 y = Number of phenolic OH group present in Morphine
 z = Number of sp^3 carbons in **Terpineol** then value of $x + y + z = ?$

2. Consider following reaction sequence



Total number of stereoisomers corresponding to product R :

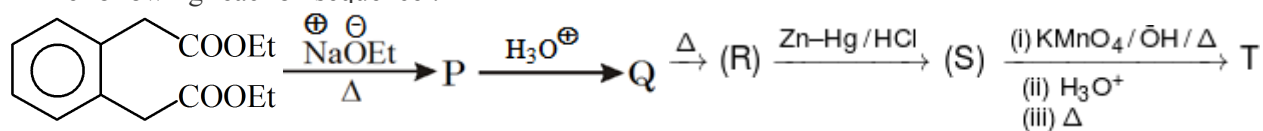
3. There is a cyclic hexapeptide (molecular weight = 488) which upon complete hydrolysis gives natural amino acids Glycine (molecular weight = 75), alanine, phenyl alanine & valine. Glycine contributes 37.8 % of total hydrolysed product. What will be total number of glycine units in this cyclic hexapeptide ?

Space for Rough Work

4. If y = Total number of compounds among the following which can undergo carbyl amine reaction.
Serotonin, Histamine, Morphine, Codeine, Cimetidine, Ranitidine, Norethindrone
Then, value of y is

5. Number of compounds give positive NaHCO_3 test.
(a) Squaric acid (b) Picric acid (c) Carbolic acid
(d) Ascorbic acid (e) Acrylic acid (f) Maleic acid
(g) Crotonic acid

6. In the following reaction sequence :



degree of unsaturation of cyclic organic product T is :

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Lot A consists of 5 good and 3 defective articles. Lot B consists of 3 good and 5 defective articles. A new Lot C is formed by taking 3 articles from A and 4 articles from B. The probability that an article chosen at random from C is defective, is :
 (A) $\frac{1}{3}$ (B) $\frac{2}{5}$ (C) $\frac{29}{56}$ (D) $\frac{29}{64}$
2. A dice is thrown thrice and the numbers obtained on each throw are a, b, c respectively and a, b, c satisfy $a^2 + 4b^2 + 4c^2 - 2ab - 4bc - 2ac = 0$
 If probability that the point (a, b, c) lies inside the tetrahedron formed by the plane $x + y + z = 10$ and coordinate planes is $\frac{6}{\lambda}$, where $\lambda \in \mathbb{N}$, then λ is
 (A) 9 (B) 12 (C) 25 (D) 27
3. Cards are drawn one by one without replacement from a well shuffled pack of 52 playing cards until 2 aces are obtained for the first time. If N is the number of cards required to be drawn, then the probability $P(N = n) = \frac{1}{k}(n-a)(n-b)(n-c)$
 where $k, a, b, c \in \mathbb{N}$ and $a > b > c$ then $a - b - c$ is equal to
 (P(N = n) is probability that n cards required to be drawn until 2 aces obtained for first time)
 (A) 1 (B) 0 (C) -1 (D) 2

Space for Rough Work

4. If the number of ways in which 5 Apples, 5 Bananas, 5 Kiwis and 5 Oranges can be distributed equally among 5 persons so that exactly 2 out of them get all 4 identical fruits and each of the remaining persons gets exactly 2 kinds of fruits, is N.
(Fruits of same species are alike)
Then, N is equal to
(A) 720 (B) 1440 (C) 2880 (D) 360
5. Consider the equation $\sqrt{1 + \frac{3 \cos 2x}{\sqrt{\sin^4 x + 4 \cos^2 x} + \sqrt{\cos^4 x + 4 \sin^2 x}}} = \sqrt{2} \sin^{-1}(\sin 2x)$, $x \in [-\pi, \pi]$
(A) Number of roots of this equation is 2 in $\left(\frac{\pi}{8}, \frac{\pi}{6}\right)$
(B) Number of roots of this equation is 4 in $\left(\frac{\pi}{8}, \frac{\pi}{6}\right)$
(C) One root lies in the interval $\left(\frac{\pi}{8}, \frac{\pi}{6}\right)$
(D) if $x \in [0, n\pi]$ $n \in \mathbb{N}$, number of roots will never be odd
6. The value(s) of x for which $\log_3(\sqrt{x} + |\sqrt{x} - 1|) = \log_9(4\sqrt{x} - 3 + 4|\sqrt{x} - 1|)$ is/ are :
(A) (0, 1) (B) $(0, 1] \cup \{4\}$ (C) (0, 1] (D) $[0, 1] \cup \{4\}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. A and B shoot independently until each shoots their target. They have probabilities $\frac{3}{5}$ and $\frac{5}{7}$ respectively of hitting the target at each shot. Then, which of the following statement(s) is/are correct ?
- (A) Probability that B requires more shots than A is $\frac{6}{31}$
- (B) Probability that B requires less shots than A is $\frac{10}{31}$
- (C) Probability that A and B require same numbers of shots is $\frac{15}{31}$
- (D) Probability that B require more shots than A is same as probability that A require more shots than B

Space for Rough Work

8. 50 identical marbles are to be distributed among four boys B_1, B_2, B_3 and B_4 . The number of marbles received by them in the distribution are as follows :

$$B_1 : 1, 3, 5, 7, \dots$$

$$B_2 : 4, 6, 8, 10, \dots$$

$$B_3 : 5, 7, 9, 11, \dots$$

$$B_4 : 2, 4, 6, 8, \dots$$

Then which of the following statement(s) is/are correct ?

- (A) Total number of ways of distribution is ${}^{22}C_3$
 (B) Total number of ways of distribution is ${}^{20}C_3$
 (C) If B_4 is receiving not more than 14 marbles, then no. of ways of distribution is 960
 (D) If B_4 is receiving not more than 14 marbles, then no. of ways of distribution is 1085

9. Let x, y, z be real numbers with $x \geq y \geq z \geq \frac{\pi}{12}$ such that $x + y + z = \frac{\pi}{2}$

(A) Maximum value of $\cos x \cdot \sin y \cdot \cos z$ is $\frac{1 + \sqrt{3}}{8}$

(B) Maximum value of $\cos x \cdot \sin y \cdot \cos z$ is $\frac{2 + \sqrt{3}}{8}$

(C) Minimum value of $\cos x \cdot \sin y \cdot \cos z$ is $\frac{1}{16}$

(D) Minimum value of $\cos x \cdot \sin y \cdot \cos z$ is $\frac{1}{8}$

Space for Rough Work

10. If n^{th} term of the series $6 + 17 + 34 + 57 + \dots$ is $(\log_2 a)n^2 + (\log_3 (b - a))n + \log_4 c$ then
- (A) $a + b = 5$
- (B) $b - c = 13$
- (C) $\log_5(2c + b) = \log_5(a + b)$
- (D) $\log_2(2c + a) = 4$
11. Which of the following statement is/are true ?
- (A) $\sum_{r=0}^n \frac{(r+3)}{(r+1)} ({}^nC_r) = 2^n + \frac{2(2^{(n+1)} - 1)}{(n+1)}$
- (B) If λ is the value of sum of coefficients in the expansion of $(3x + 2y + 4z)^{50}$, then the last digit of λ is 1
- (C) $1 + \sum_{r=1}^2 ({}^2C_r) + \sum_{r=1}^3 ({}^3C_r) + \dots + \sum_{r=1}^{20} ({}^{20}C_r) = (2^{21} - 22)$
- (D) $\sum_{r=1}^{10} r(r!) is divisible by 2$
12. The average value of $2\sin 2^\circ, 4\sin 4^\circ, 6\sin 6^\circ, \dots, 180\sin 180^\circ$ is
- (A) $\tan 89^\circ$ (B) $\tan 45^\circ$ (C) 0 (D) $\cot 1^\circ$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. The number of 5 digit numbers that can be formed using the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 in which one digit appears once and two digits appear twice (eg., 44533) is k, then the value of $\frac{k}{1890}$ is equal to
 2. If sum of the values of x satisfying the equation

$$\sqrt{2} \sin\left(x + \frac{\pi}{4}\right) (\sec^2 x - (1 + \sqrt{3}) \tan x + 2 + \sqrt{3}) = \sqrt{9 + 18 \sin x \cos x}$$
 in $[0, 2\pi]$ is $\left(\frac{p}{q}\right) \pi$, where p and q are co-prime, then $|p - q|$ is equal to
-

Space for Rough Work

3. If the least positive value of x satisfying the equation $\frac{\sin x}{\cos 3x} + \frac{\sin 3x}{\cos 9x} + \frac{\sin 9x}{\cos 27x} = 0$ is $\frac{\pi}{k}$ where $k \in \mathbb{N}$ then $\left(\frac{k}{5}\right)$ is equal to
4. If $\frac{\cos x + \cos y + \cos z}{\cos(x+y+z)} = 2$ and $\frac{\sin x + \sin y + \sin z}{\sin(x+y+z)} = 2$ then the value of $\cos(x+y) + \cos(y+z) + \cos(z+x)$ is equal to :
(where $x, y, z \in \mathbb{R}$)
5. Let $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$ where $a_0, a_1, a_2, \dots, a_{20}$ are constant then the value of $\frac{a_7}{a_{13}}$ is equal to
6. Let a_k denotes coefficient of x^k in expansion of $(1 + 2x)^n$, $x \in \mathbb{N}$.
If $\sum_{k=0}^n (3k + 1)a_k = (pn + q)r^n$, $p, q, r \in \mathbb{N}$
then $p + q + r$ is equal to

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work